Can missing data teach protein dynamics?

Dyna-1

Learning millisecond protein dynamics from what is missing in NMR spectra

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Nature **658**, 1077–1087 (2026)

DOI: 10.1038/s41586-026-10989-4

Journal club

Sequence + structure → residue exchange propensity

The question: where does a protein change state?

Prediction target
Identify residues that experience changes in chemical environment on the microsecond–millisecond timescale.

- Such exchange can accompany catalysis, ligand binding and allostery.
- A single structure does not specify exchange timescales.
- Direct, standardized dynamics measurements are scarce: the authors curate 133 proteins, far too few for broad supervised learning.

Opportunity: chemical-shift assignment databases are much larger.

Introduction; Fig. 1a,b. Exchange can also arise from binding or oligomerization.

Two meanings of a protein being dynamic

Fast local motion: ps–ns
Backbone flexibility changes heteronuclear NOE and relaxation. Low hetNOE often identifies these motions.

Conformational exchange: μs–ms
Interconversion changes chemical shifts and can add an exchange contribution R_{ex} to transverse relaxation:

$$R_2 = R_2^0 + R_{\mathrm{ex}}.$$

R_2 is a **rate**, measured in s^{-1}. Faster decay means a broader NMR peak.
The same residue can experience motion on both timescales.

Fig. 1a,d; Methods. These categories are not mutually exclusive.

Core idea: a peak can disappear because states exchange



Weak supervision
Train on "assignment missing", then test whether the learned score predicts *measured exchange* in independent relaxation datasets.

Missingness is an imperfect proxy: overlap, incomplete assignment, isotope labelling and solvent exchange can also remove signals. Assigned residues may still exchange.

Fig. 2a,b; main text, "Curating proxy labels for exchange. Schematic is explanatory.

Three datasets separate learning from validation

Dataset	Scale	Role and observable
mBMRB	9,381 curated entries	Training labels: backbone ^{15}N assignment present or missing.
RelaxDB	133 proteins	Transfer test: R_1, R_2, hetNOE plus assignment information.
RelaxDB-CPMG	10 proteins	Direct relaxation-dispersion test; complete raw curves for six.

The model never trains on RelaxDB exchange labels.

Prospective experiments on two held-out proteins provide an additional evidence tier.

Figs. 1–3, 5–6; journal Data availability and author repository.

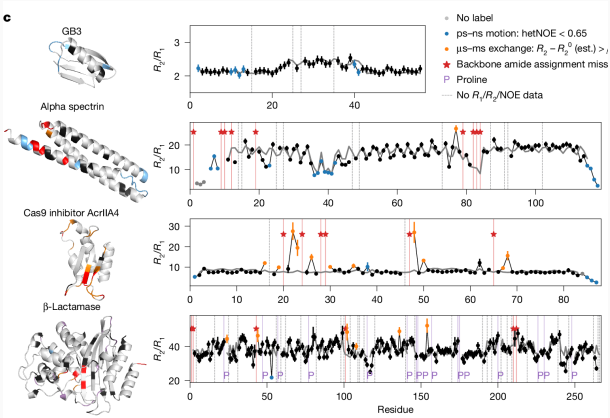
Building mBMRB requires removing trivial missingness

- Start from 12,102 polypeptide entities in the BMRB (May 2023 query).
- Remove deuterated or partial/methyl-labelled samples, ambiguous sequences and insufficiently assigned entries; mask terminal His tags.
- Exclude entries with ≥ 15 consecutive internal missing assignments to reduce unassigned-domain artefacts.
- Exclude prolines and unassigned termini from evaluation. Proline lacks a backbone amide hydrogen.

Residual noise
Missingness still depends on temperature, pH, protein size and assignment practice. The label is not a direct kinetic measurement.

Methods, "The mBMRB dataset; journal Extended Data Fig. 2.

RelaxDB labels account for how the whole protein tumbles



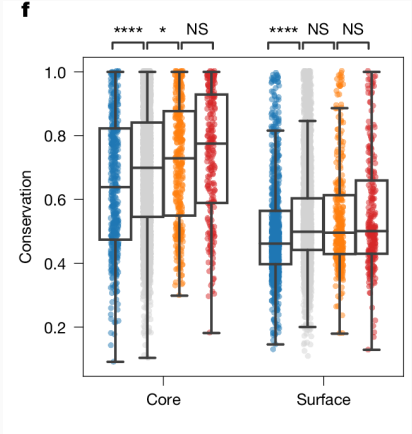
High R_2/R_1 can reflect **anisotropic tumbling**, not exchange.

1. Predict a structure with AF2.
2. Calculate rigid-tumbling relaxation with HYDRONMR.
3. Fit a protein-specific scale.
4. Label excess relaxation beyond experimental and fitting uncertainty.

Blue: fast motion. Orange: exchange. Red stars: missing assignments.

Fig. 1c (cropped). Grey curves: rigid-tumbling estimates; black points: measured ratios.

An evolutionary clue motivates language-model features



Residues with measured μs–ms exchange are more conserved than residues with no motion label.

Fast-motion residues are less conserved. Missing-assignment residues resemble the exchange group.

Inference: evolutionary sequence information may help predict functionally relevant exchange.

Conservation alone does not establish that every missing peak reflects exchange.

Fig. 1f (cropped); colours follow Fig. 1: blue fast, grey no label, orange exchange, red missing.

Dyna-1: frozen ESM-3 features, trained residue classifier



- Best version combines sequence and structure; sequence-only and structure-only variants are also evaluated.
- Backbone model weights remain fixed; train the head to classify missing assignments.
- Compare to ESM-2, AF2 pair features, amino-acid/secondary-structure/SASA models, normal modes and conservation.

Fig. 3b; journal Extended Data Fig. 4; Methods, "Deep learning training. Diagram adapted.

A toy example: the output is a propensity, not a movie

Residue	Ala	Gly	Ser	Pro	Leu
Assignment label	present	missing	present	masked	present
Illustrative score	0.10	0.85	0.70	—	0.15

- Gly provides a positive training example; Ser is treated as negative even if it has detectable exchange.
- A high Ser prediction could therefore be useful dynamics information rather than a biological false positive.
- The score does not specify the alternate structure, exchange rate, state populations or motion amplitude.

These numbers are invented solely to explain the learning task.

Toy illustration; task definition in Figs. 2–4. No experimental values shown.

A score labelled probability needs careful interpretation

The authors adjust logits for the rarity of missing assignments. For raw logit ℓ and missing-class prior $\pi_m \approx 0.05$:

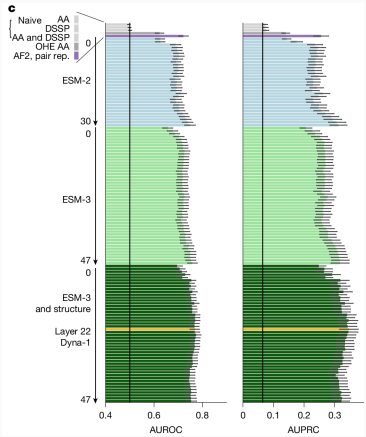
$$p_{\text{balanced}} = \sigma\left(\ell - \log \frac{\pi_m}{1 - \pi_m}\right).$$

- This targets **balanced classification**, rather than the observed prevalence of missing assignments.
- A raw logit near -3 becomes a balanced score near 0.5.
- The monotonic adjustment does not change residue ranking or AUROC.

Treat $p(\text{exchange})$ as a prioritization score; it is not an experimentally calibrated probability under arbitrary conditions.

Journal Supplementary Information 2, Supplementary Fig. 2. Final sentence: journal-club interpretation.

Model selection: deeper features help, the last layer need not win



In the initial, more permissive split:

ESM-3 + structure, layer 22:
AUROC 0.77.

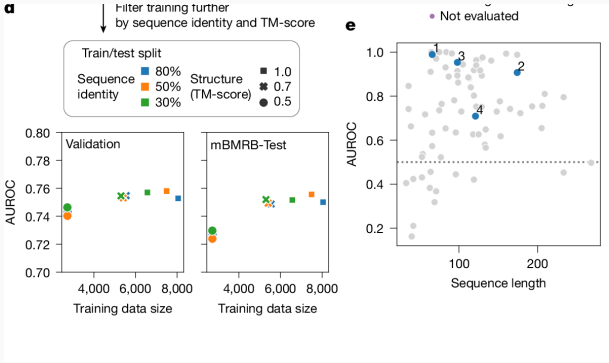
ESM-2 last layer: 0.75.
AF2 pair representation: 0.71.

AUPRC helps distinguish models when positives are rare.

The best hidden representation is an *intermediate* layer, not simply the final output.

Fig. 3c (cropped); journal main text, p. 1081. Error bars: 95% CIs across proteins.

Homology control makes the test harder



Reserve mBMRB test proteins, RelaxDB and RelaxDB-CPMG before training.

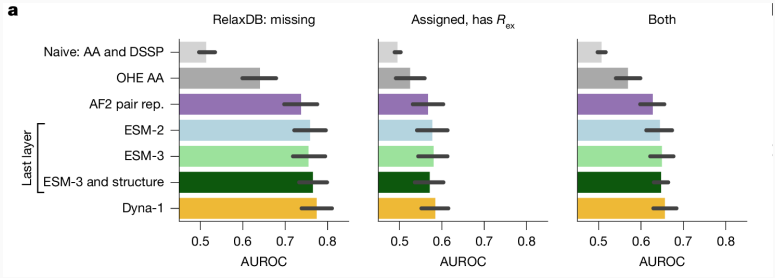
Final training uses **30% sequence identity** and **TM-score 0.5** filtering.

Training shrinks to roughly 2,700 proteins; validation AUROC falls to about **0.74**.

This supports transfer beyond close homologues, while not eliminating every possible pretraining overlap.

Fig. 3d,e (cropped); Methods, data splitting. Final caveat is our interpretation.

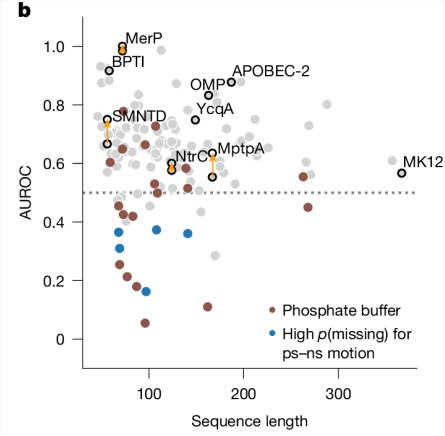
The decisive transfer test excludes missing assignments



- Left: missing assignments; middle: **assigned residues with measured** R_{ex}; right: both.
- The middle task cannot be solved merely by recalling missingness labels.
- Dyna-1 retains predictive power, but performance decreases substantially: the proxy is informative and incomplete.

Fig. 4a (cropped), n=112 proteins after phosphate-related exclusions; error bars: 95% CIs.

Average performance hides strong context dependence



Mean RelaxDB AUROC: **0.63** across all 133 proteins; **0.66** after excluding phosphate-related datasets.

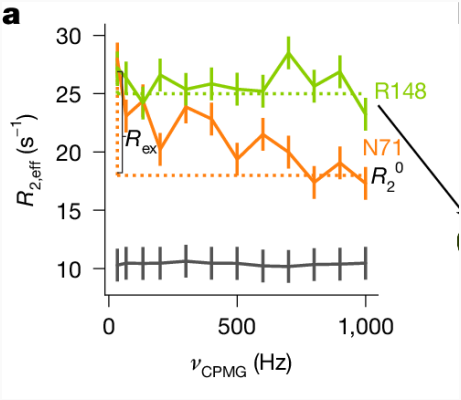
Some poorly predicted exchange occurs in phosphate-binding proteins measured in phosphate buffer.

Other failures involve fast-motion regions.

Interpretation: reporting both the full and curated benchmark is essential; exclusions followed examination of failures.

Fig. 4b (cropped); Extended Data Figs. 3b, 6; journal p. 1083. Dotted line: random ranking.

CPMG asks a sharper question about exchange



CPMG varies the pulse repetition frequency ν_{CPMG} (Hz).

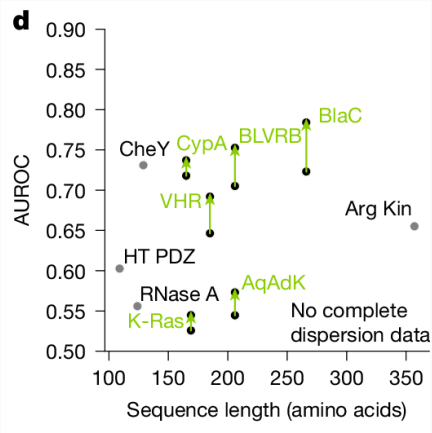
Orange N71: faster pulses suppress exchange, reducing $R_{2,\mathrm{eff}}$ (s^{-1}).

Green R148: the curve stays elevated. A difference between low and high frequency can miss exchange that remains unsuppressed.

Dyna-1 flagged R148 in cyclophilin A.

Fig. 5a (cropped); Methods, "Identifying unsuppressed Rex. A flat elevated curve needs a baseline.

Reanalysis recovers signal, but depends on a physical baseline



Estimate exchange-free R_2 using scaled HYDRONMR rigid-tumbling predictions.

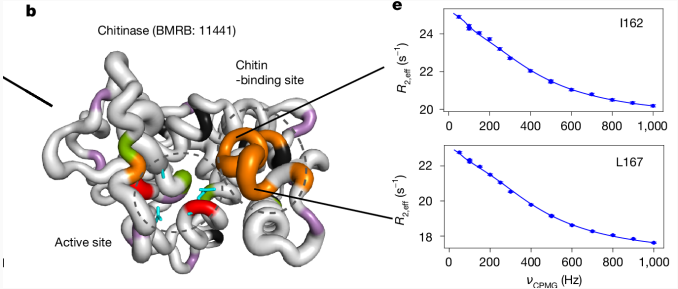
Include residues with elevated, unsuppressed R_{ex} as well as conventional dispersion.

The journal caption reports higher AUROC in **5 of 6** reprocessed proteins. Four others lacked complete raw curves.

Caution: revised labels were motivated by predictions, so these gains need prospective support.

Fig. 5d (cropped); journal Fig. 5 caption: 5/6; main text p. 1085 inconsistently states all six.

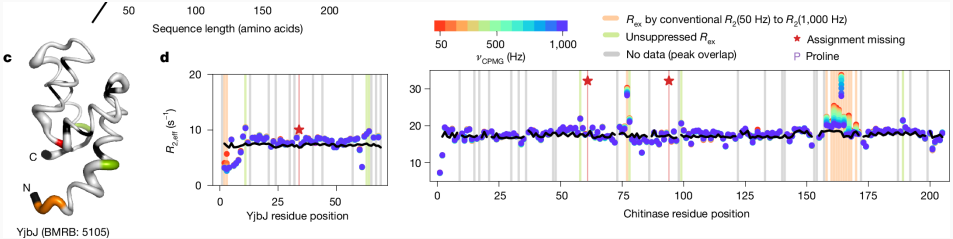
Prospective test: choose high- and low-exchange proteins



- Choose two proteins from held-out mBMRB-Test after model release.
- High predicted exchange: *Bryum coronatum* GH19 chitinase (BMRB 11441).
- Low predicted exchange: hypothetical protein YjbJ (BMRB 5105).
- Measure ^{15}N CPMG at 900 MHz; this tests new dynamics data, not new assignment labels.

Fig. 6b,e (cropped); journal Methods, NMR spectroscopy. Selected extremes are not a random prospective cohort.

Prospective results support regional predictions, modestly



- Chitinase: exchange in active-site and chitin-binding regions; residue AUROC **0.62**.
- YjbJ: conventional dispersion at residues 2 and 3; residue AUROC **0.72**. Additional green labels denote unsuppressed exchange.
- Stronger evidence than retrospective relabelling, but only two selected proteins, with few positives in YjbJ.

Fig. 6c,d (cropped); journal p. 1086. Black curve: estimated rigid baseline; shading as Fig. 5.

What this paper establishes, and what remains open

Supported

Missing assignments contain transferable information about μs–ms exchange.

Large, noisy experimental archives can supervise useful predictors.

Functional regions can be prioritized for further dynamics experiments.

Open

Rates, state populations, alternate structures and trajectories remain unpredicted.

Temperature, pH, ligands and sidechains affect the observables.

Training proteins are under 400 residues; large assemblies and other regimes need evaluation.

Discussion: what would a large, blinded prospective benchmark need to measure to distinguish learned exchange from general flexibility?

Discussion; journal Extended Data Fig. 6. Benchmark proposal is journal-club interpretation.

Sources and a practical way to use Dyna-1

Primary citation: Wayment-Steele et al., *Nature* 658, 1077–1087 (2026). doi.org/10.1038/s41586-026-10989-4

Reading basis: full final journal article and Methods, all main figures and extended-data captions, and journal Supplementary Information 1–3. Material claims and metrics were reconciled against the supplied 30-page journal PDF.

Code and data: github.com/WaymentSteeleLab/Dyna-1; relaxation archives at Zenodo 20834076.

Practical interpretation: run sequence + structure, inspect high-scoring regions, and test them with experiments under the biological conditions of interest. Use the score to select questions; retain all assay metadata.

Sources, scientific qualifications, figure provenance and build QA are recorded in the accompanying README.